CO2-AT1

Analytical problem solving

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Machine Learning.

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1) Given:-
Study hours (x_1) = 2
Attendance (x_2) = 1
weights: $w_1 = 0.6$, $w_2 = 0.4$
Bias (b) = -0.5

Sol:-
Net input = $w_1x_1 + w_2x_2 + b$.
$= (0.6 \times 2) + (0.4 \times 1) - 0.5$
$= 1.2 + 0.4 - 0.5$
$= 1.1$

Step activation function.

$$y(net) = \begin{cases} 1, & net \geq 0 \\ 0, & net < 0 \end{cases}$$

∴ The perceptron output is 1, indicating the student passes.

2) Given:-
Initial weight (w) = 0.5
Learning rate (η) = 0.1
Input (x) = 2.
Actual output (y) = 0.6
Target output (t) = 1.

Sol:-
output error = $t - y$.
$= 1 - 0.6$
Error = 0.4

Weight update

$\Delta w = \eta(t-y)x$

$= 0.1(1-0.6)2$

$= 0.08$

New weight

$w_{new} = w_{old} + \Delta w$

$= 0.5 + 0.08$

$w_{new} = 0.58$

3) Given:-

True positive (TP) = 180

True Negative (TN) = 150

False positive (FP) = 20

False Negative (FN) = 50.

Sol:-

Total predictions

$N = 180 + 150 + 20 + 50$

$N = 400$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

$$= \frac{180 + 150}{400} \times 100$$

$= 82.5\%$

$$\text{precision} = \frac{TP}{TP + FP} \times 100 = \frac{180}{180 + 20} \times 100$$

$= 90\%$

$$= \frac{TP}{TP+FN} \times 100$$

$$= \frac{180}{180+50} \times 100$$

$$= 78.26\%.$$

$$\text{Specificity} = \frac{TN}{TN+FP} \times 100$$

$$= \frac{150}{150+20} \times 100$$

$$= 88.24\%.$$

∴ Accuracy = 82.5%.
Precision = 90%.
sensitivity = 78.26%.
specificity = 88.24%.

4) Given:-
Initial population = 2, 4, 6, 8

sol:-
Fitness Function.

Individual(x)	Binary representation	Fitness (x^2)
2	0010	4
4	0100	16
6	0110	36
8	1000	64

- Best individual : $x = 8$, fitness = 64
- Next best : $x = 6$, fitness = 36.

∴ 6 and 8 selected for cross-over

Parent 1 : 8 = 1000
Parent 2 : 6 = 0110

Offspring 1 = 1000 = 8.
Offspring 2 = 0110 = 6

Role of mutation :-

Mutation randomly changes one or more bits in an individual chromosome. It introduces new genatic variation, maintain population diversity. helps explore new solutions.

5) Given :-
- Training accuracy increases 60% to 95%.
- validation accuracy initially increases from 58% to 72%, then decreases to 68%.

sol :-

- Issue :-

The model is experiencing overfitting training accuracy continues to improve while validation accuracy decreases after reaching its peak.

- Techniques to prevent overfitting:

1) Dropout: Randomly disables some of the neurons during training, reducing excessive dependence on particular neurons and improving generalization.

2) Early stopping: Monitors validation performance and stops training when validation accuracy stops improving or validation loss begins to increase.